

FR1504E Rear Helper Spring Kit Installation Manual

Model: FR1504E (Rear Suspension Package)

Please read these instructions carefully before commencing installation. Ensure all standard safety protocols are observed when working beneath the vehicle.

1. Parts List

Before beginning, verify that all components listed in the table below are present in your kit.

Ref #	Description / Specifications	Qty
1	Weldment Bracket (Left-Hand & Right-Hand)	1L - 1R
2	Rubber Spacer (4" O.D. x 1" Thick with AH Bushing)	2
3	Hollow Rubber Spring	2
4	Hex Cap Screw (HCS) M10 x 40 x 1.5, Grade 8.8, Zinc Plated	2
5	Hex Cap Screw (HCS) M10 x 65 x 1.5, Grade 8.8, Zinc Plated	2
6	Split Lock Washer M10, Grade 8.8, Zinc Plated	2
7	Hex Cap Screw (HCS) M10 x 30 x 1.5, Grade 8.8, Zinc Plated	2
8	Axle Support Weldment	2
9	Saddle Clamp 3-1/2"	2
10	Flat Washer 3/8" x 1", Grade 2, Zinc Plated	4
11	Nylock Locknut 3/8" UNC, Grade 5, Zinc Plated	4

2. Installation Instructions

1. Raise Vehicle: Elevate the vehicle using a suitable lift or jack, allowing the rear axle to articulate freely. Ensure the chassis is securely supported before proceeding.

2. Remove OEM Bump Stop: Remove the existing original equipment manufacturer (OEM) bump stop assembly from the vehicle frame.

3. Assemble Spring and Bracket: Integrate the hollow rubber spring (Ref #3) with the upper weldment bracket (Ref #1) as depicted in the assembly diagram.

4. Mount Upper Assembly: Secure the assembled upper bracket and spring to the vehicle frame using the existing mounting holes and the provided fasteners (Ref #4, #6, #7).

5. Install Lower Support: Attach the lower axle support weldment (Ref #8) and saddle clamp (Ref #9) onto the vehicle axle using the provided fasteners (Ref #10, #11). Tighten the nuts until a snug fit is achieved.

6. Lower Vehicle: Carefully lower the vehicle to the ground.

Installation Notes:

- Verify that brake lines or brake cables do not interfere with the installed rubber spring. Adjust routing if necessary.

- If necessary, a rubber spacer (Ref #2) and a longer hex cap screw (Ref #5) can be utilized to achieve the desired ride height.
- Note: Illustrations are for reference only. If no specific reference image is provided, please install according to the standard mechanical guidelines.

3. Clearance Instructions & Fitting Guidance

- Clearance Gap: For rear helper spring kits installed on light-duty trucks and vans, an optimal clearance of 1/2 inch to 1-1/2 inches should be maintained between the rubber spring and its contact point when the vehicle is unladen.
- Load Effects: The presence of a partial load, cargo, or aftermarket vehicle accessories will reduce or eliminate this initial clearance gap.
- Heavy-Duty Applications: For medium and heavy-duty trucks, trailers, and motorhomes, the helper springs are designed to be in constant contact with no discernible gap.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended axle weight rating.
- Inspect for potential interference between the hollow rubber spring and other vehicle components. Under load, the rubber spring will experience an increase in diameter.
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